

Previous Year Questions

Some Basic Concepts of Chemistry

Short Answer (2 Marks)

Q1: Define molar volume and state its value at STP. (2 Marks)

- Molar volume is the volume occupied by one mole of a gas at standard temperature and pressure (STP).
- Its value is 22.4 L mol^{-1} at STP (273.15 K and 1 atm).

Q2: What is the difference between empirical formula and molecular formula? (2 Marks)

- Empirical formula gives the simplest whole number ratio of atoms in a compound.
- Molecular formula gives the actual number of atoms of each element in one molecule.

Q3: Define limiting reagent with an example. (2 Marks)

- The reactant that gets completely consumed first in a chemical reaction and limits the amount of product formed is called the limiting reagent.
- Example: In the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, if 2 mol of H_2 and 1 mol of O_2 are taken, H_2 is the limiting reagent.

Q4: What is meant by significant figures? (2 Marks)

- Significant figures are the digits in a number that carry meaningful information about its precision.
- They include all certain digits and the first uncertain digit.

Q5: State Avogadro's law. (2 Marks)

- Avogadro's law states that equal volumes of all gases at the same temperature and pressure contain the same number of molecules.
- Mathematically: $V \propto n$ (at constant T and P).

Q6: Define atomic mass unit (amu). (2 Marks)

- One atomic mass unit (amu) is defined as one twelfth the mass of a carbon-12 atom.

- $1 \text{ amu} = 1.6605 \times 10^{-24} \text{ g}$.

Q7: What do you mean by stoichiometry? (2 Marks)

- Stoichiometry is the quantitative study of reactants and products in a balanced chemical equation.
- It allows calculation of masses or volumes of substances involved in reactions.

Q8: Define mole. (2 Marks)

- A mole is the amount of substance that contains as many entities (atoms, molecules, ions, etc.) as there are in 12 g of carbon-12.
- One mole contains 6.022×10^{23} entities (Avogadro's number).

Q9: What is the law of multiple proportions? (2 Marks)

- When two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other are in simple whole-number ratios.

Q10: Define molarity and write its unit. (2 Marks)

- Molarity is the number of moles of solute per liter of solution.
- Unit: mol L^{-1} .

Medium Answer (3 Marks)

Q11: Explain the law of conservation of mass with an example. (3 Marks)

- The law of conservation of mass states that mass can neither be created nor destroyed in a chemical reaction.
- Example: In the reaction $2H_2 + O_2 \rightarrow 2H_2O$, the total mass of reactants equals the total mass of products.
- This law forms the basis for balancing chemical equations.

Q12: Differentiate between accuracy and precision. (3 Marks)

- *Accuracy* refers to how close a measured value is to the true value.
- *Precision* refers to how close repeated measurements are to each other.

- A measurement can be precise but not accurate, and vice versa.

Q13: Calculate the number of molecules in 4.4 g of CO_2 . (3 Marks)

- Molar mass of $CO_2 = 12 + 2 \times 16 = 44 \text{ g mol}^{-1}$
- Moles of $CO_2 = \frac{4.4}{44} = 0.1 \text{ mol}$
- Number of molecules $= 0.1 \times 6.022 \times 10^{23} = 6.022 \times 10^{22}$ molecules

Q14: State and explain Dalton's atomic theory. (3 Marks)

- Dalton proposed that matter is made up of indivisible atoms.
- Main postulates include:
 - - Atoms are indivisible and indestructible.
 - - All atoms of an element are identical.
 - - Atoms combine in simple whole-number ratios to form compounds.
 - - Chemical reactions involve rearrangement of atoms.

Q15: How is empirical formula calculated from percentage composition? (3 Marks)

- Convert percentage to mass (assume 100 g sample).
- Convert mass to moles by dividing by atomic masses.
- Divide each mole value by the smallest to get simplest ratio.
- If necessary, multiply to get whole numbers — this gives the empirical formula.

Q16: A sample of compound contains 0.24 g of carbon and 0.64 g of oxygen. Find the empirical formula. (3 Marks)

- Moles of C $= \frac{0.24}{12} = 0.02$
- Moles of O $= \frac{0.64}{16} = 0.04$
- Ratio = C:O $= 0.02 : 0.04 = 1 : 2$
- Empirical formula $= CO_2$

Q17: Calculate the molarity of solution containing 5 g of $NaOH$ in 250 mL of solution. (3 Marks)

- Molar mass of $NaOH = 40 \text{ g mol}^{-1}$

- Moles of $NaOH = \frac{5}{40} = 0.125 \text{ mol}$
- Molarity = $\frac{0.125}{0.250} = 0.5 \text{ mol L}^{-1}$

Q18: What are the advantages of using SI units in measurements? (3 Marks)

- SI units provide a standard system used globally, ensuring consistency.
- They are based on decimal system making calculations easier.
- SI units can be used in all branches of science and engineering.

Q19: What are the basic postulates of the atomic mass theory? (3 Marks)

- Atoms have definite masses.
- Atomic masses of elements are constants.
- Atoms combine in fixed ratios to form compounds.

Q20: Explain the law of definite proportions. Give an example. (3 Marks)

- This law states that a chemical compound always contains the same elements in the same proportion by mass.
- Example: Water (H_2O) always has hydrogen and oxygen in a mass ratio of 1:8.

Long Answer (5 Marks)

Q21: Derive the relationship between molecular formula and empirical formula. (5 Marks)

- Let empirical formula mass = EFM
- Let molecular formula = $n \times$ empirical formula
- Given molar mass (MM), $n = \frac{MM}{EFM}$
- Once n is calculated, multiply subscripts in empirical formula by n to get the molecular formula.
- Example: If empirical formula = CH_2O , molar mass = 180 g/mol , $EFM = 30 \text{ g/mol}$
- Then $n = \frac{180}{30} = 6$
- So molecular formula = $C_6H_{12}O_6$

Q22: Discuss in detail the various laws of chemical combinations with suitable examples. (5 Marks)

- *Law of Conservation of Mass*: Total mass remains constant. Example: Combustion of methane.
- *Law of Definite Proportions*: Compounds have constant element ratios. Example: Water is always H_2O .
- *Law of Multiple Proportions*: Two elements can form different compounds with mass ratios being whole numbers.
- *Gay-Lussac's Law*: Gases combine in volume ratios of small whole numbers. Example: $2H_2 + O_2 \rightarrow 2H_2O$ (volumes).
- *Avogadro's Law*: Equal volumes of gases at STP contain equal molecules.

Q23: Calculate the empirical and molecular formula of a compound containing 40

- Assume 100 g compound:
- - C = 40 g $\rightarrow 40/12 = 3.33\text{mol}$ - H = 6.7g $\rightarrow 6.7/1 = 6.7\text{mol}$
- - O = 53.3 g $\rightarrow 53.3/16 = 3.33\text{mol}$ Divide by smallest :
- - C = 1, H = 2, O = 1
- Empirical formula = CH_2O , EFM = 30
- $n = \frac{180}{30} = 6$, so Molecular formula = $C_6H_{12}O_6$

Q24: Describe the steps involved in the calculation of percentage composition from a molecular formula. (5 Marks)

- Step 1: Write molecular formula.
- Step 2: Calculate molar mass by summing atomic masses.
- Step 3: For each element:
- % of element = $\frac{\text{mass of element in 1 mol}}{\text{molar mass}} \times 100$
- Step 4: Repeat for each element.

Q25: Explain the method of determining limiting reagent in a chemical reaction. (5 Marks)

- Step 1: Write balanced chemical equation.

- Step 2: Calculate moles of each reactant.
- Step 3: Use stoichiometry to find required moles.
- Step 4: Compare available moles to required; the one that runs out first is the limiting reagent.

Q26: A reaction has 5.0 g of $CaCO_3$ reacting with 10.0 g of HCl. Determine the limiting reagent and calculate the mass of $CaCl_2$ formed. (5 Marks)

- Equation: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
- Moles of $CaCO_3 = \frac{5.0}{100} = 0.05 \text{ mol}$
- Moles of HCl = $\frac{10.0}{36.5} \approx 0.274 \text{ mol}$
- Required HCl = $0.05 \times 2 = 0.1 \text{ mol} < 0.274 \text{ mol} \rightarrow CaCO_3$ is limiting reagent
- Moles of $CaCl_2 = 0.05 \text{ mol} \rightarrow Mass = 0.05 \times 111 = 5.55 \text{ g}$

Q27: What are the basic SI units used in chemistry? Explain their importance. (5 Marks)

- SI Units in Chemistry:

- - Length: metre (m)
- - Mass: kilogram (kg)
- - Time: second (s)
- - Amount: mole (mol)
- - Temperature: kelvin (K)

- Importance:

- - Standardized measurements
- - Consistent communication
- - Ease of conversions and calculations

Q28: Discuss the concept of mole and its significance in chemistry. (5 Marks)

- Mole is a counting unit, like dozen.
- $1 \text{ mole} = 6.022 \times 10^{23} \text{ entities}$
- Helps in expressing quantities of atoms, molecules, ions in measurable terms



- Allows stoichiometric calculations
- Connects microscopic (atoms) to macroscopic (grams) world

Q29: Explain different types of chemical reactions with examples. (5 Marks)

- *Combination*: $H_2 + Cl_2 \rightarrow 2HCl$
- *Decomposition*: $2H_2O \rightarrow 2H_2 + O_2$
- *Displacement*: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
- *Double displacement*: $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
- *Redox*: Oxidation and reduction happen simultaneously

Q30: What is meant by stoichiometry? Illustrate with a solved numerical problem. (5 Marks)

- Stoichiometry is the calculation of quantities in chemical reactions based on balanced equations.
- Example:
- Reaction: $2Mg + O_2 \rightarrow 2MgO$
- How many grams of MgO formed from 12 g Mg ?
- Moles of $Mg = \frac{12}{24} = 0.5$
- Moles of $MgO = 0.5$ (1:1 ratio)
- Mass = $0.5 \times 40 = 20$ g

